LAB 07

FROM ASSUMPTIONS TO EXPERIMENT DESIGN

Identify • Prioritize • Hypothesize • Design • Review

TODAY'S GOAL Starting from your venture and the early learning from Lab 06, select ONE riskiest assumption, formulate a testable hypothesis, and design an experiment with observable metrics and pre-defined success criteria.

MANDATORY DURING LAB 07 Complete ONE experiment-design cycle and submit ONE team PDF on D2L before leaving the lab. Include the assumption map, selected assumption, hypothesis, experiment protocol, success criteria, and peer feedback. This checkpoint contributes to LM2.

1. Lab Plan

Phase	Time	Core activity	Output
IDENTIFY	25 min	List assumptions and existing evidence	Assumption inventory
PRIORITIZE	30 min	Select the riskiest assumption	Risk justification
HYPOTHESIZE	30 min	Define an observable prediction	Testable hypothesis
DESIGN	45 min	Choose experiment and predefine metrics	Experiment plan
REVIEW & SUBMIT	20 min	Peer review, revise and submit	D2L checkpoint

THE LAB IN ONE LINE VENTURE → ASSUMPTIONS → RISKIEST ASSUMPTION → TESTABLE HYPOTHESIS → EXPERIMENT → SUCCESS CRITERIA → CHECKPOINT

The plan structures approximately 2h30 of guided work. Remaining Lab Hours may be used for feedback, preparing an experimental artifact, and continuing venture work.

2. Phase 1 — IDENTIFY

Reconnect with Labs 05 and 06. Your MVP and prototype tests may have generated early learning, but many business and product assumptions remain untested.

START WITH THE VENTURE An assumption is something you believe must be true for the venture to work. Distinguish what you believe from what you have actually observed.

Venture / Software Product and target users

Our venture is a university event discovery platform for UDST. The idea is to have one organized place where students and faculty can see upcoming university events with the main details they need (date, time, location, organizer, purpose, and if there is food or a certificate), plus basic search and filters. Our target users are current UDST students and faculty, and event organizers are secondary users.

What did you learn in Lab 06? What remains uncertain?

In Lab 06 we tested our prototype with a few students and faculty. They were able to browse the events, use the filters and find the event details without much help, so the main flow seems easy to use. But the test only showed that they CAN use it when we ask them to. We are still not sure if students and faculty would actually open it by themselves in real life, because Navigate360 already has an events section and most people we asked don't use it.

Assumption inventory

Category	Important assumption	Current evidence (strong / weak / none)
Desirability	A1: UDST students and faculty will go back to the event platform on their own (without reminders) to check upcoming events.	Weak: students and faculty say they want one place for events, but most of them don't use Navigate360 which already exists.
Desirability	A2: Students and faculty miss events mainly because they find out too late, not because they are not interested.	Strong: most people in our survey said they missed events because they found out too late.

Feasibility	A3: We can collect events from Outlook, Navigate360 and club social media and keep the platform updated on time.	None: we haven't tried collecting or updating events regularly yet.
Viability	A4: UDST administration and IT would be willing to support or adopt the platform.	None: we haven't talked to the university or IT about it yet.

BE SPECIFIC Avoid broad statements such as “users like our app.” State who will do what, in which situation, and why that behavior matters.

OUTPUT 1 — An inventory of venture assumptions and the evidence currently supporting them.

3. Phase 2 — PRIORITIZE & MAP

Map your assumptions using importance and existing evidence. Then choose ONE high-impact, weakly supported assumption to test first. If several qualify, discuss which assumptions other decisions depend on. Do not automatically select the easiest to test.

RISKIEST ASSUMPTION High importance × weak evidence. Explain both dimensions rather than treating this expression as a numerical formula.

MONITOR High importance · Strong evidence Assumption IDs: A2	**TEST FIRST** High importance · Weak / no evidence Assumption IDs: A1, A3
SET ASIDE Low importance · Strong evidence Assumption IDs: None	**INVESTIGATE LATER** Low importance · Weak / no evidence Assumption IDs: A4

ASSUMPTION MAP — Importance: high (top) → low (bottom); existing evidence: strong (left) → weak / none (right). Place the IDs of your assumptions in the quadrants. Discuss dependencies before selecting ONE to test first.

Selected riskiest assumption

A1: UDST students and faculty will go back to the event platform on their own (without us reminding them) to check what events are coming up.

Why would the venture be at risk if this assumption is false?

Our whole idea depends on students and faculty actually visiting the platform. If they only open it once and forget about it, then putting all events in one place doesn't really fix anything and they will keep missing events like before. Also the other assumptions depend on this one: organizers won't care about posting events (A3) and the university won't support it (A4) if no one is using it. That's why we chose A1 to test first and not the others.

What evidence do you already have, and why is it insufficient?

From our survey, most people said they missed events because they found out too late, and they said they would like one place for events. But this is only what they say, not what they actually do. The evidence we have actually points the other way: Navigate360 is already a central place for events and most students and faculty don't know about it or don't use it. Our prototype test also only showed that the flow is usable, not that they would come back to it by themselves.

ONE ASSUMPTION ONLY During the guided Lab 07 cycle, design one experiment for the selected assumption. Other important assumptions can be addressed later as part of LM2.

OUTPUT 2 — One prioritized, justified riskiest assumption.

4. Phase 3 — HYPOTHESIZE

Transform the assumption into a prediction that could be contradicted by observations. Define what people will actually do, not merely what they say they like.

HYPOTHESIS PATTERN We believe that [specific target users] will [observable behavior] because [reason]. We will know this when [metric + threshold + time window].

Who are the target users? In what context?

Current UDST students and faculty who have missed at least one university event before because they heard about it too late. The context is a normal teaching week during the semester, when event emails get mixed with all the course emails in Outlook.

What observable behavior would support the assumption?

Students and faculty open the event platform again by themselves on different days after their first visit, without us sending any reminder or message.

Why do you expect this behavior?
Because checking one list is easier than searching through Outlook, club Instagram pages and asking friends. People in our survey also told us they regret missing events, so we think they have a reason to check.

Your complete hypothesis

We believe that UDST students and faculty who have missed university events before will go back to a central event listing on different days to check upcoming events, because it saves them from searching in Outlook, social media and asking friends. We will know this when at least 60% of the participants (6 out of 10) return to the listing on 2 or more separate days within 14 days of their first visit, without getting any reminders from us.

CHECK TESTABILITY Could the experiment produce a result that contradicts your hypothesis? If not, revise the prediction. Interest or satisfaction alone is not necessarily a meaningful commitment.

What observation would count against your hypothesis?
If most participants open the platform once and never come back (the same thing that happens with Navigate360). Also if at the end they tell us they still heard about events mainly from Outlook or friends, even though the events were on our listing.

OUTPUT 3 — One complete, falsifiable, observable and time-bound hypothesis.

5. Phase 4 — DESIGN

Select the simplest ethical and practical experiment that can produce evidence about your hypothesis. Choose the experiment because of the learning goal, not because a tool is fashionable.

What you need to learn	Possible experiment
Problem occurrence / existing behavior	Past-behavior interviews
Interest or initial commitment	Landing page / early-access signup / fake door
Value delivered in real use	Concierge pilot / targeted prototype test
Technical feasibility	Time-boxed technical spike

Chosen experiment and why it fits the hypothesis
We chose a concierge pilot. We will publish our prototype as a live website and for two weeks we will add real upcoming UDST events to it manually (we collect them ourselves from Outlook, Navigate360 and club social media). This fits because our hypothesis is about real return behavior, so we need students and faculty using it with real events, not just saying they like it. A survey or landing page would only measure interest, and we already have that.

Experiment component	Your plan
Participants / recruitment	10 participants: 7 UDST students from different majors and years, and 3 faculty members. We will recruit students in person on campus and through class group chats, and faculty by asking our instructors and emailing departments. We ask one screening question: "Have you ever missed a university event because you found out too late?" and only include people who say yes. We will try not to use only our close friends.
Participant action / task	On day 1 we send them the link and explain the platform in about 1 minute. After that we don't contact them at all, they just use it (or not) like they normally would.
Artifact or setup needed	A live version of our prototype with search and filters. We update it with real UDST events every 2 to 3 days. Each event shows date, time, location, organizer, purpose, food/certificate tags and a link to the original announcement.
Observable metric and collection method	Main metric: % of participants who come back on 2 or more separate days after day 1. We track this with a simple analytics tool and give each participant a unique link code, so we don't store names or emails. Extra: on day 14 we ask one short question: "Where did you first hear about the last event you looked at?"
Duration / time window	14 days during a normal teaching period (we will avoid exam weeks).
Risks, consent, or ethical safeguards	We ask for consent when recruiting and people can leave anytime. No personal data is collected, only anonymous link codes. The website clearly says it's a student course project and not an official UDST service. Every event links to its original source so the info stays correct. We won't send reminders because that would affect the result.

Predefine interpretation before execution

Outcome	Predefined criterion
SUCCESS: evidence supporting the hypothesis	60% or more of participants (6 or more out of 10) come back on 2 or more separate days within the 14 days.
FAILURE: evidence against the hypothesis	30% or less (3 or fewer out of 10) come back on 2 or more days. This would mean students and faculty act the same as with Navigate360, and we would need to rethink our direction (for example sending events where people already are, like Outlook or Instagram).
INCONCLUSIVE: insufficient / ambiguous evidence	Only 4 or 5 out of 10 come back, or fewer than 8 participants stay in the pilot for the full 14 days, or the tracking doesn't work and we lose data.

DESIGN BEFORE RESULTS Specify metrics and thresholds before running the test. A landing page or signup form is an option, not a requirement. Do not claim validation from a peer review or a hypothetical experiment.

OUTPUT 4 — One executable experiment protocol with pre-defined decision criteria.

6. Phase 5 — REVIEW & SUBMIT

Exchange plans with another team. Review the logic of the experiment, not the attractiveness of its presentation. Feedback during the lab is a design check, not evidence validating your market hypothesis.

Peer review question	Feedback / revision
Is the assumption genuinely important and weakly supported?	Yes. The other team agreed that if students and faculty don't come back to the platform the whole idea fails, and our evidence for it is weak since Navigate360 is not used.
Is the hypothesis observable and falsifiable?	Mostly yes, because returning visits can be counted and the result can go against us. They suggested we clearly define what counts as a "visit".
Will the experiment measure the proposed behavior?	They said it measures return visits, but the first few days might just be curiosity because the site is new, not a real habit.
Are thresholds, participants, and timing defined beforehand?	Yes, the thresholds and 14 days are set before running it. Their concern was that 10 people recruited through our classes might be biased in our favour.

What will you change in your experiment design after peer feedback?

We will only count visits from day 3 onwards as real "returns" so the first few days of curiosity don't make the result look better. We will also try to recruit at least half of the participants from outside our own classes and friend groups.

FROM LAB 07 TO LM2 The guided lab requires ONE complete experiment design. After the lab, prepare the necessary artifact, conduct the experiment with relevant participants when feasible, document actual observations, and address additional high-risk assumptions. Preserve the original criteria when reporting outcomes. Lab 08 will focus on interpreting evidence.